

Emerging Technologies: An Overview, Key Areas, and Strategic Insights for Technology Adoption

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I. Introduction

Technology continues to change the way people live, study, work, and communicate. As new ideas and inventions develop, they can make everyday tasks easier and create new ways to solve problems. These new and developing technologies are known as **Emerging Technologies**. Emerging technology refers to new or developing tools, systems, and ideas that can greatly affect people and society. Some examples include artificial intelligence, robotics, virtual reality, and renewable energy technologies. Although some of these technologies are still being improved, they are already becoming part of different areas such as education, business, healthcare, and communication. A technology may also be emerging when an existing innovation continues to progress over time, even after a decade, and is applied creatively to new areas. One example of an emerging technology is Artificial Intelligence (AI); it was first developed in the 1950s, but it remains emerging because it continues to advance and is used more widely across industries. Learning about emerging technology matters because it helps us understand how the world is changing and how these technologies may affect our future. It also helps us see both the benefits and potential challenges of using new technology.

This comprehensive report is intended to serve as a strategic reference for business and government stakeholders. It explains the concept of emerging technologies, identifies the key factors used to recognize them, examines major areas of technological innovation, and discusses their contribution to the development of the Fourth Industrial Revolution. The report was prepared to provide useful insights that can support client recommendations, informed decision-making, and strategic planning.

II. Characteristics of an Emerging Technology

According to Rotolo, Hicks, and Martin (2015), there are five (5) main characteristics of an emerging technology that help explain how technologies emerge. These are **Radical Novelty**, **Relatively Fast Growth**, **Coherence**, **Prominent Impact**, and **Uncertainty** and **Ambiguity**.

Table 1. Characteristics of an Emerging Technologies		
Characteristics	Description	Example
Radical Novelty	A technology can count as radically novel even if it is not a brand new, so long as it solves a problem using a fundamentally different approach than earlier solutions, or is applied to an entirely new purpose.	Artificial intelligence being repeatedly redirected toward new uses like generating more AI video contents and solving a lot of problems, even though the underlying field is decades old.
Relatively Fast Growth	Emerging technologies expand more quickly than established ones. This growth can be measured through rising numbers of	The increasing number of individuals from companies, universities, and private and public sectors adopting AI tools.

	contributors, increased funding, and an increasing number of research output, prototypes, and commercial products.	
Coherence	Previously unrelated fields of research start to merge into something unified, and with a clear internal logic connecting them.	Cloud + AI because cloud computing provides the storage and computing power needed by AI to process large amounts of data. While AI analyzes data, identifies patterns, and so on.
Prominent Impact	Refers to the significant and noticeable effects that can have on people, organizations, industries, or society. It considers a prominent impact in emerging technology when it largely changes the way people live, work, communicate, or perform tasks.	Artificial Intelligence automates tasks and assists in a lot of things across business, education, healthcare, and everyday lives.
Uncertainty and Ambiguity	Uncertainty and ambiguity mean that the future effects of an emerging technology are difficult to predict and understand because the technology is still developing and may be interpreted or used differently by different groups.	Experts can predict the use of AI in the future, but its success and impact on jobs remain uncertain because some people see AI as a tool that improves work, while others fear it may replace certain jobs.

2.1 Conclusion About the Characteristics

To conclude the five characteristics of an emerging technology, it is also important to remember that technology everyday is still being developed. It always improves and changes quickly over time, and its capabilities, limitations, and possible risks are still being studied. The five emerging technologies described above means that the technology is introduced in new approaches, develops fast, connects different fields of study, and has the potential to create significant social and economic impacts. It is still surrounded by uncertainty since there are no time machine yet to view the future of technology. However, this uncertainty does not mean that the technology is weak or ineffective. It is a natural part of innovation. The most important effects of emerging technologies may only become clear as they continue to develop and be adopted in the future.

III. Emerging Technology Areas

There are areas in emerging technology that are new, rapidly developing, and have the potential to change how people view reality, solve problems quickly, and work easily for things conveniently. Among the important emerging technology areas are **Augmented Reality (AR)**, **Artificial Intelligence (AI)** and **Machine Learning**, and **Quantum Computers**. These areas are expected to have the potential and capabilities to change manual or traditional practices of certain fields in the industry and in every life.

Table 2. Emerging Technology Areas		
Emerging Technology Areas	Definition	Example
Augmented Reality	Digital content overlays the view of the real world.	• A car-in display - shows driving and parking guides. • Pokemon Go - places pokemon in the real world. • Furniture Apps - Shows how furniture looks in your room.
Artificial Intelligence	A machine that mimics the behavior, voice, and intelligence of humans.	• ChatGpt • Siri • Application Assistant tools
Machine Learning	Computers learning from the collected data.	• In healthcare, machine learning can help quickly identify and develop new medicines.
Quantum Computers	Computers using quantum principles.	• Google develops quantum computers, including the Willow chip, to advance quantum computing research through its Quantum AI program.

3.1 Augmented Reality (AR) - a technology that overlays digital contents, such as images, information, sound, and 3D objects, onto the top of a person's view of the physical world. It allows users to see virtual or digital elements while seeing the actual world around them.

3.2 Artificial Intelligence (AI) - AI is a machine that enables a device to perform tasks quickly that normally require human input and knowledge. Nowadays, it can perform a lot of things, like AI's can talk through the screen, perform tasks that the users command, generate images, video contents, etc.

3.2.1 Machine Learning - it is a specific application within AI. It is the mechanism that allows a machine to improve through experience in a way that resembles human learning. It allows systems to learn from data they collected and improve their performance without being specifically programmed for every task.

3.3 Quantum Computers - a type of computer that uses the principles of quantum mechanics to process and store information. Unlike traditional computers that use bits represented as 1 and 0, quantum computers use quantum bits (qubits), which can represent multiple possible states. So far, some of the largest quantum computers demonstrated by research laboratories have been developed by companies such as IBM, Google, Rigetti Computing, and IonQ.

To conclude these certain areas in emerging technology, it's important to remember that these areas of technologies are rapidly developing and can have potential risks to us, like people may rely on technology too much, and possible threats like technology can be attacked or misused. However, these areas can bring many benefits and opportunities as it makes tasks easier and faster, improves productivity, etc. The new challenges and risks they have needs to be managed responsibly.

IV. The Fourth Industrial Revolution (4IR)

Technological advancements have changed how products are made and how industries operate. These changes happened through different stages known as the **Industrial Revolution**, where production became more advanced by combining physical machines with modern technologies. These developments helped improve the quality of the products, reduce costs, and make work more efficient, while also changing people's jobs, working conditions, and daily lives.

Table 3. The Four Main Stages of the Industrial Revolution, Including Their Time Periods and Focus.		
Industry	Period	Focus
Industry 1.0	18th Century	• More on mechanical productions • Hydroelectric and steam-powered equipment.
		• Development of Electricity supply

Industry 2.0	19th Century	• Steel and oil industries • More advanced machinery
Industry 3.0	20th Century	• Electronics • Early computers • Integrated circuit
Industry 4.0	21st Century — ongoing	• Artificial Intelligence (AI) • Augmented Reality (AR) • Cloud computing • Internet of things (IoT) • Robotics • Virtual Reality (VR)

4.1 First Industrial Revolution (Industry 1.0)

The first industrial revolution began in the 18th century. Introduced mechanical production powered by hydroelectric and steam-based equipment, marking the shift away from artisan or manual, hand-based production. It marked the beginning of organizational structures and urbanization, which changes lifestyles from agrarian to industrial.

4.2 Second Industrial Revolution (Industry 2.0)

The Second Industrial Revolution began in the 19th century and was marked by the development of electricity and the growth of the steel and oil industries. Electricity was introduced into factories and production systems, allowing for increased production and the creation of more advanced machinery.

4.3 Third Industrial Revolution (Industry 3.0)

The Third Industrial Revolution marked the beginning of the digital age. Computers, the internet, electronics, and information technology transformed manufacturing and communication by increasing automation and connectivity, while the use of renewable energy sources continued to grow.

4.4 Fourth Industrial Revolution (Industry 4.0)

The Fourth Industrial Revolution and the present period of technology is emerging as a period of rapid, technology-driven change in the 21st century. It is powered by innovations such as AI, robotics, quantum computing, IoT, VR, and AR, transforming how businesses operate and how suppliers, producers, customers, and people interact with technology.

V. Selected Building Blocks of the Fourth Industrial Revolution or Industry 4.0

The Fourth Industrial Revolution (4IR) is built on the combination of digital, physical, and biological technologies. Its key building blocks include **Big Data Analytics**, the **Internet of Things (IoT)**, **Cloud Computing**, and **Augmented Reality (AR)**. These technologies show how industries continue to evolve over time by offering more advanced features that improve efficiency and user convenience.

5.1 Big Data Analytics - It involved the practice of gathering, organizing, and analyzing huge amounts of data from different sources to identify useful patterns, trends, and insights. It helps organizations make faster and more informed decisions by turning large datasets into meaningful information for areas like production, operations, and customer management.

5.2 Internet of Things (IoT) - Internet of Things (IoT) is a technology concept where physical devices are connected to the internet, allowing them to collect, send, receive, and exchange data. These devices typically use sensors, software, and network connections to monitor information and perform specific actions. For example, a smartwatch can track a user's steps and heart rate, then send this information to an app for monitoring and analysis.

5.3 Cloud Computing - This refers to the delivery of computing services such as storage, software, and processing power through the internet instead of relying only on a local computer. As cloud technology becomes faster and more reliable, data can be processed almost instantly like the reaction times can reach several milliseconds, allowing businesses to store and analyze more machine data in the cloud and support more efficient, data-driven production systems.

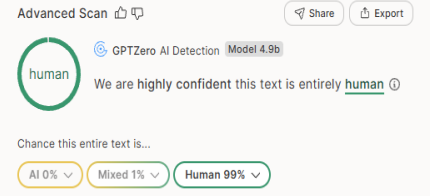
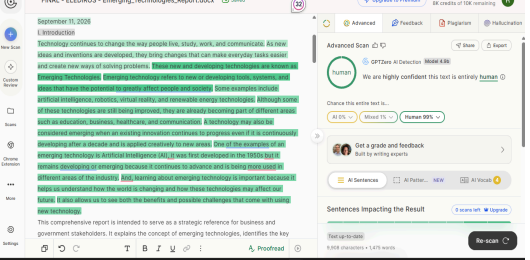
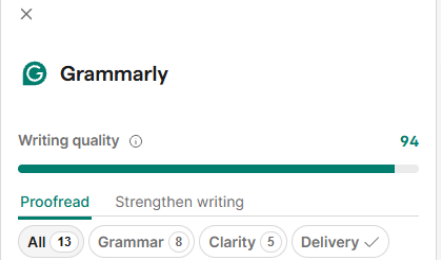
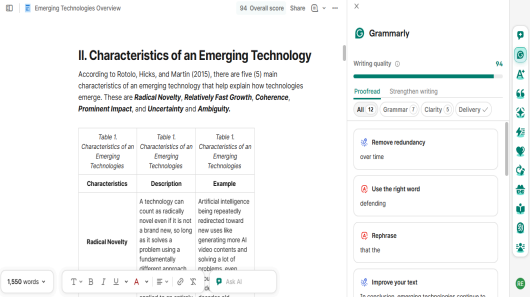
5.4 Augmented Reality (AR) - In an industrial setting, AR can help workers perform tasks more easily, like helping warehouse staff find parts or providing technicians with step-by-step repair instructions through a mobile device. While AR is still developing in the industry, it is expected to become more widely used in the future for faster decision-making and more efficient work processes.

VI. Conclusion

In conclusion, emerging technologies continue to develop and improve over time, and have a lot of uses depending on what type of technology that the users use. The characteristics of emerging technology such as the radical novelty, fast growth, coherence, prominent impact, and uncertainty and ambiguity explain why some technologies continue to be viewed as emerging even after their initial development. The emerging technology areas show how emerging technologies can influence different activities and industries. At the same time, the four stages of the industrial revolution show the different periods of changes in the field of technology. It also shows progression shifting from mechanical to production to increasingly digital, connected, and intelligent production environments. Industry 4.0 represents the current and developing stage, where selected technologies from industry 4.0 contribute to changes in production and in the relationships among people, machines, and organizations.

To conclude this comprehensive report, it's always important to remember that studying emerging technologies is not only about understanding new inventions and innovations in technology, but also about recognizing their impact on our lives, education, businesses, and society. Emerging technologies continue to develop and can provide many opportunities for growth, efficiency, and improvement. However, they also come with challenges and risks that require responsible use, critical thinking, and awareness. So, understanding these technologies helps us prepare for the future and make informed decisions about how they should be developed and used for the benefit of society.

VII. AI Checker and Grammarly Results

Checker	Result	Proof
AI Checker	 Advanced Scan human GPTZero AI Detection Model 4.9b We are highly confident this text is entirely human Chance this entire text is... AI 0% Mixed 1% Human 99% AI Detector All clear! Nothing in your text appears to be AI generated No AI-generated text detected 0% Does it seem accurate? Recommended steps Moderate AI patterns 1 addressed	 FINAL - ELEDIROS - Emerging_Technologies_Report.docx September 11, 2025 Introduction Technology continues to change the way people live, study, work, and communicate. As new ideas and inventions are developed, they can make everyday tasks easier and create new ways to solve problems. These new and developing technologies are known as Emerging Technologies. Emerging technology refers to new or developing tools, systems, and ideas that can greatly affect people and society. Some examples include artificial intelligence, robotics, virtual reality, and renewable energy technologies. Although some of these technologies are still being improved, they are already becoming part of different areas such as education, business, healthcare, and communication. A technology may also be considered emerging when an existing innovation continues to progress even if it is continuously developing after a decade and is applied creatively to new areas. One of the examples of an emerging technology is Artificial Intelligence (AI). It was first developed in the 1950s but it remains developing as emerging technology continues to advance and is being more used in different areas of the industry. Artificial intelligence is important because it helps us understand how the world is changing and how these technologies may affect our future. It also allows us to see both the benefits and possible challenges that come with using new technology. This comprehensive report is intended to serve as a strategic reference for business and government stakeholders. It explains the concept of emerging technologies, identifies the key Emerging Technologies Overview 99 Overall score AI Detector All clear! Nothing in your text appears to be AI generated No AI-generated text detected 0% Does it seem accurate? Recommended steps Moderate AI patterns 1 addressed Before you turn it in AI Detector Strip out phrases that may appear to be AI-generated
Grammarly	 Grammarly Writing quality 94 Proofread Strengthen writing All 13 Grammar 8 Clarity 5 Delivery	 Emerging Technologies Overview 94 Overall score Grammarly Writing quality 94 Proofread Strengthen writing All 13 Grammar 8 Clarity 5 Delivery Remove redundancy over time Use the right word defending Rephrase that the Improve your text To maximize readability and ensure clarity

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